

Jiamao Liu

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EDUCATION

Xiamen University

M.S., Bioinformatics, Average Score: 85/100

Xiamen, China

Sep. 2024 – Present

Coursework: Scientific Writing (89), Molecular and Cellular Biology (87), Frontiers in Life Sciences Research (96)

Xiamen University

B.A., Biological Science, Average Score: 80/100

Xiamen, China

Sept. 2020 – Jun. 2024

Coursework: Biostatistics (97), Cell and Microscope Experiment (94), AI Application and Practice (92)

RESEARCH INTERESTS

Single-cell and spatial multi-omics; gene regulatory elements and enhancers; epigenomics; deep learning

PUBLICATIONS

Co-first authors; * Corresponding author

Published

- Hong, D.#, Shu, M.#, **Liu, J.#**, et al. (2025). Divergent combinations of enhancers encode spatial gene expression. *Nature Communications*, 16, 5091.

In Revision

- Zhu, M.#, Zha, Z.#, Xuan, J., Zhao, H., Dai, J., **Liu, J.**, et al. Super-CUT&Tag: A Sensitive, Spatially Resolved Approach for Epigenomic Profiling in Tissue Sections. *bioRxiv* (2025).

Under Review

- Shu, M.#,*, **Liu, J.#**, Wang, M.#, Chen, Z.#, et al. Decoding the spatiotemporal and epigenetic principles underlying human spinal cord development.

RESEARCH EXPERIENCE

Deep Learning for Spatial Epigenomics and Enhancer–Gene Regulatory Inference

Jun. 2026 – Present

Project Lead, Ongoing project

- Developing a deep learning framework to reconstruct spatial epigenomic landscapes from spatial transcriptomics and paired single-cell multiome data and infer spatial enhancer–gene regulatory relationships.
- Developing representation learning strategies to jointly model gene expression, chromatin accessibility, and spatial context, capturing nonlinear and spatially heterogeneous regulatory patterns beyond conventional correlation-based approaches.
- Contributions: established the spatial RNA–multiome integration and spatial ATAC inference pipeline; developed metrics for spatial regulatory complexity; currently designing deep learning models for regulatory pattern discovery and gene prioritization.

Non-invasive Embryo Assessment through cfDNA Methylation Profiling

Nov. 2025 – Present

Independent computational project

- Developing a computational framework to identify pregnancy outcome-associated biomarkers from cfDNA methylation in spent embryo culture medium (SECM) for non-invasive embryo assessment.
- Integrated quality control, maternal DNA decontamination, multi-scale methylation profiling, differential analysis, and CNV analysis for low-input and sparsely covered cfDNA data.

- Contributions: established the analysis workflow; implemented DECENT/NICE-based decontamination strategies; currently identifying candidate biomarkers and developing predictive models for embryo quality assessment.

Multi-omics Atlas of Human Spinal Cord Development

Mar. 2024 – Present

Co-first author, manuscript under submission

- This project constructs a developmental multi-omics atlas of the human spinal cord by integrating 74,000+ single-cell transcriptomes, chromatin accessibility profiles, and spatial transcriptomics datasets across seven developmental stages.
- We identified lineage-specific enhancer-gene regulation, chromatin priming events, transcription factor dynamics, developmental trajectories, and spatially resolved disease-risk regulatory programs.
- Contributions: led computational analysis; designed enhancer-gene linkage, chromatin priming, trajectory, footprinting, and spatial GWAS analyses; generated most figures and computational results; contributed substantially to manuscript writing and submission.

Spatial GWAS Enrichment Analysis for Tissue Epigenomics

Jun. 2025 – Nov. 2025

Independent computational project

- This project extends disease-variant enrichment analysis from single-cell chromatin accessibility data to spatial epigenomic profiles.
- The analysis demonstrated that spatial epigenomic data can improve localization of disease-associated regulatory programs within anatomically defined tissue regions.
- Contributions: developed the spatial GWAS enrichment workflow; performed statistical analysis and visualization; evaluated disease-risk localization across spatial tissue sections.

Spatial Enhancer Code Reveals Regional Gene Regulation

Sept. 2023 – Jun. 2025

Co-first author, published in **Nature Communications** (2025)

- This study investigated how combinations of enhancers encode spatially restricted gene expression patterns during tissue development.
- By integrating spatial transcriptomics and epigenomic profiling, we identified a spatial enhancer code in which distinct enhancer combinations cooperatively determine regional transcriptional specificity.
- Contributions: developed computational methods for enhancer-combination analysis; performed data integration, statistical analysis, and figure generation; contributed to manuscript revision.

AWARDS

- National Scholarship for Graduate Students, China (Top 3%, 2025)
- Outstanding Student Award, Xiamen University (Top 5%, 2025)
- First Prize Poster Award, 5th Annual Symposium of the Fujian Bioinformatics Society (2025)
- Software Copyright: *eSpatial: Decoding Spatial Regulatory Mechanisms of Enhancers* (Third Contributor, 2025)

SKILLS

Programming: Python, R, PyTorch, Linux, High-Performance Computing, Git, Docker

Bioinformatics: Scanpy, Seurat, SnapATAC2, ArchR, Signac, SCENIC+, Bismark, deepTools, Cursor, Codex

Languages: Chinese (Native), English (Professional Working Proficiency)